10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Review Exercise 3 — Full Solutions & MCQs

Unit 3 | Review Exercise | New Book (PECTAA)

Review Exercise 3 (Solved)

1. Four possible answers are given for the following questions. Choose the correct answer.

(i) If the two matrices given below are equal, then a =

$$\begin{bmatrix} a+2 \\ 3 \end{bmatrix} = \begin{bmatrix} 7 \\ 3 \end{bmatrix}$$

(a) 3 (b) 5

(c) 6 (d) 7

Answer: (b) 5 — equal matrices have equal corresponding elements, so $a + 2 = 7 \Rightarrow a = 5$.

(ii) A is a _____ matrix.

$$A = \begin{bmatrix} 3 & 5 & 0 \end{bmatrix}$$

(a) row (b) square

(c) column (d) null

Answer: (a) row — it has one row and three columns, so it is a row matrix.

(iii) B is a _____ matrix.

$$B = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

(a) identity (b) square

(c) row (d) column

Answer: (d) column — it has one column and two rows, so it is a column matrix.

(iv) M is a/an _____ matrix.

$$M = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(a) rectangular (b) identity

(c) column (d) row

Answer: (b) identity — the diagonal elements are 1 and all the other elements are 0.

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(v) If $A^t = -A$, then A is a _____ matrix.

(a) symmetric (b) row
(c) rectangular (d) skew-symmetric

Answer: (d) skew-symmetric — a matrix which is equal to the negative of its transpose is called skew-symmetric.

(vi) If A = [3 4] and B = [7 8], then A + B =

$$A + B = \begin{bmatrix} 3 & 4 \end{bmatrix} + \begin{bmatrix} 7 & 8 \end{bmatrix} = \begin{bmatrix} 10 & 12 \end{bmatrix}$$

(a) [21 32] (b) [24 28]
(c) [10 12] (d) [11 11]

Answer: (c) [10 12] — add the corresponding elements: 3 + 7 = 10 and 4 + 8 = 12.

(vii) If A = [3 4] and B = [7 8], then B − A =

$$B - A = \begin{bmatrix} 7 & 8 \end{bmatrix} - \begin{bmatrix} 3 & 4 \end{bmatrix} = \begin{bmatrix} 4 & 4 \end{bmatrix}$$

(a) [10 12] (b) [11 11]
(c) [4 4] (d) [−4 −4]

Answer: (c) [4 4] — 7 − 3 = 4 and 8 − 4 = 4.

(viii) What is the additive inverse of [[3], [4]]?

$$-\begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} -3 \\ -4 \end{bmatrix}$$

(a) [[−3], [4]] (b) [[−3], [−4]]
(c) [[3], [4]] (d) [[−1], [4]]

Answer: (b) [[−3], [−4]] — we change the sign of every element, because A + (−A) = O.

(ix) For the matrix A given below, the order of A is

$$A = \begin{bmatrix} 3 & 1 & 3 \\ 2 & 0 & 2 \end{bmatrix}$$

(a) 3-by-2 (b) 2-by-3
(c) 3-by-3 (d) 2-by-2

Answer: (b) 2-by-3 — A has 2 rows and 3 columns, so its order is 2 × 3.

(x) Find the value of |3 1; 0 4|

$$\begin{vmatrix} 3 & 1 \\ 0 & 4 \end{vmatrix} = (3)(4) - (1)(0) = 12$$

(a) 11 (b) 12
(c) 13 (d) −11

Answer: (b) 12 — |A| = (3)(4) − (1)(0) = 12 − 0 = 12.

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2. If A = [[4, 2], [7, 6]] and B = [[5, 5], [8, 8]], then find:

$$A = \begin{bmatrix} 4 & 2 \\ 7 & 6 \end{bmatrix} \quad B = \begin{bmatrix} 5 & 5 \\ 8 & 8 \end{bmatrix}$$

(i) $(A - B)^t$ (ii) $B^t - A^t$ (iii) $2A + 3B$

(i) $(A - B)^t$

$$A - B = \begin{bmatrix} 4 & 2 \\ 7 & 6 \end{bmatrix} - \begin{bmatrix} 5 & 5 \\ 8 & 8 \end{bmatrix} = \begin{bmatrix} -1 & -3 \\ -1 & -2 \end{bmatrix}$$

Interchange rows and columns

$$(A - B)^t = \begin{bmatrix} -1 & -1 \\ -3 & -2 \end{bmatrix}$$

Answer: $(A - B)^t$ = [[−1 −1], [−3 −2]]

(ii) $B^t - A^t$

$$B^t = \begin{bmatrix} 5 & 8 \\ 5 & 8 \end{bmatrix} \quad A^t = \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}$$

$$B^t - A^t = \begin{bmatrix} 5 & 8 \\ 5 & 8 \end{bmatrix} - \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 3 & 2 \end{bmatrix}$$

Answer: $B^t - A^t$ = [[1 1], [3 2]]

(iii) 2A + 3B

$$2A = \begin{bmatrix} 8 & 4 \\ 14 & 12 \end{bmatrix} \quad 3B = \begin{bmatrix} 15 & 15 \\ 24 & 24 \end{bmatrix}$$

$$2A + 3B = \begin{bmatrix} 8 & 4 \\ 14 & 12 \end{bmatrix} + \begin{bmatrix} 15 & 15 \\ 24 & 24 \end{bmatrix} = \begin{bmatrix} 23 & 19 \\ 38 & 36 \end{bmatrix}$$

Answer: 2A + 3B = [[23 19], [38 36]]

3. If A = [[2, 3], [4, 5]], B = [[5, 6], [2, 1]] and C = [[3, 2], [1, 7]], then verify:

$$A = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} \quad B = \begin{bmatrix} 5 & 6 \\ 2 & 1 \end{bmatrix} \quad C = \begin{bmatrix} 3 & 2 \\ 1 & 7 \end{bmatrix}$$

(i) $2(A + B) = 2A + 2B$

(ii) $(A + B) + C = A + (B + C)$

(iii) $(A + B)C = AC + BC$

(iv) $C(A - B) = CA - CB$

(v) $(AB)^{-1} = B^{-1}A^{-1}$

(vi) $AA^{-1} = A^{-1}A = I$

(vii) $(AB)^t = B^tA^t$

(viii) $(AB)C = A(BC)$

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(i) 2(A + B) = 2A + 2B

$$A + B = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} + \begin{bmatrix} 5 & 6 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 9 \\ 6 & 6 \end{bmatrix}$$

$$\text{L.H.S.} = 2 \begin{bmatrix} 7 & 9 \\ 6 & 6 \end{bmatrix} = \begin{bmatrix} 14 & 18 \\ 12 & 12 \end{bmatrix}$$

$$2A = \begin{bmatrix} 4 & 6 \\ 8 & 10 \end{bmatrix} \quad 2B = \begin{bmatrix} 10 & 12 \\ 4 & 2 \end{bmatrix}$$

$$\text{R.H.S.} = 2A + 2B = \begin{bmatrix} 14 & 18 \\ 12 & 12 \end{bmatrix}$$

Answer: L.H.S. = R.H.S. (verified)

(ii) (A + B) + C = A + (B + C)

$$B + C = \begin{bmatrix} 5 & 6 \\ 2 & 1 \end{bmatrix} + \begin{bmatrix} 3 & 2 \\ 1 & 7 \end{bmatrix} = \begin{bmatrix} 8 & 8 \\ 3 & 8 \end{bmatrix}$$

$$\text{L.H.S.} = \begin{bmatrix} 7 & 9 \\ 6 & 6 \end{bmatrix} + \begin{bmatrix} 3 & 2 \\ 1 & 7 \end{bmatrix} = \begin{bmatrix} 10 & 11 \\ 7 & 13 \end{bmatrix}$$

$$\text{R.H.S.} = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} + \begin{bmatrix} 8 & 8 \\ 3 & 8 \end{bmatrix} = \begin{bmatrix} 10 & 11 \\ 7 & 13 \end{bmatrix}$$

Answer: L.H.S. = R.H.S. (verified)

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(iii) (A + B)C = AC + BC

$$A + B = \begin{bmatrix} 7 & 9 \\ 6 & 6 \end{bmatrix}$$

$$\text{L.H.S.} = \begin{bmatrix} 7 & 9 \\ 6 & 6 \end{bmatrix} \times \begin{bmatrix} 3 & 2 \\ 1 & 7 \end{bmatrix} = \begin{bmatrix} 30 & 77 \\ 24 & 54 \end{bmatrix}$$

$$AC = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} \times \begin{bmatrix} 3 & 2 \\ 1 & 7 \end{bmatrix} = \begin{bmatrix} 9 & 25 \\ 17 & 43 \end{bmatrix}$$

$$BC = \begin{bmatrix} 5 & 6 \\ 2 & 1 \end{bmatrix} \times \begin{bmatrix} 3 & 2 \\ 1 & 7 \end{bmatrix} = \begin{bmatrix} 21 & 52 \\ 7 & 11 \end{bmatrix}$$

$$\text{R.H.S.} = AC + BC = \begin{bmatrix} 9 & 25 \\ 17 & 43 \end{bmatrix} + \begin{bmatrix} 21 & 52 \\ 7 & 11 \end{bmatrix}$$

$$= \begin{bmatrix} 30 & 77 \\ 24 & 54 \end{bmatrix}$$

Answer: L.H.S. = R.H.S. (verified)

(iv) C(A − B) = CA − CB

$$A - B = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} - \begin{bmatrix} 5 & 6 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} -3 & -3 \\ 2 & 4 \end{bmatrix}$$

$$\text{L.H.S.} = \begin{bmatrix} 3 & 2 \\ 1 & 7 \end{bmatrix} \times \begin{bmatrix} -3 & -3 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} -5 & -1 \\ 11 & 25 \end{bmatrix}$$

$$CA = \begin{bmatrix} 3 & 2 \\ 1 & 7 \end{bmatrix} \times \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 14 & 19 \\ 30 & 38 \end{bmatrix}$$

$$CB = \begin{bmatrix} 3 & 2 \\ 1 & 7 \end{bmatrix} \times \begin{bmatrix} 5 & 6 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 19 & 20 \\ 19 & 13 \end{bmatrix}$$

$$\text{R.H.S.} = CA - CB = \begin{bmatrix} 14 & 19 \\ 30 & 38 \end{bmatrix} - \begin{bmatrix} 19 & 20 \\ 19 & 13 \end{bmatrix}$$

$$= \begin{bmatrix} -5 & -1 \\ 11 & 25 \end{bmatrix}$$

Answer: L.H.S. = R.H.S. (verified)

(v) $(AB)^{-1} = B^{-1}A^{-1}$

$$AB = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} \times \begin{bmatrix} 5 & 6 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 16 & 15 \\ 30 & 29 \end{bmatrix}$$

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$$|AB| = \begin{vmatrix} 16 & 15 \\ 30 & 29 \end{vmatrix}$$

$= (16)(29) - (15)(30) = 464 - 450$

$= 14$

$$\text{adj } (AB) = \begin{bmatrix} 29 & -15 \\ -30 & 16 \end{bmatrix}$$

$$(AB)^{-1} = \frac{1}{14}\begin{bmatrix} 29 & -15 \\ -30 & 16 \end{bmatrix}$$

For B: $|B| = (5)(1) - (6)(2)$

$= 5 - 12$

$= -7$

$$\text{adj B} = \begin{bmatrix} 1 & -6 \\ -2 & 5 \end{bmatrix} \quad B^{-1} = \frac{-1}{7}\begin{bmatrix} 1 & -6 \\ -2 & 5 \end{bmatrix}$$

For A: $|A| = (2)(5) - (3)(4)$

$= 10 - 12$

$= -2$

$$\text{adj A} = \begin{bmatrix} 5 & -3 \\ -4 & 2 \end{bmatrix} \quad A^{-1} = \frac{-1}{2}\begin{bmatrix} 5 & -3 \\ -4 & 2 \end{bmatrix}$$

$$B^{-1}A^{-1} = \frac{1}{14}\begin{bmatrix} 1 & -6 \\ -2 & 5 \end{bmatrix} \times \begin{bmatrix} 5 & -3 \\ -4 & 2 \end{bmatrix}$$

First multiply the matrices, then multiply by 1/14

$$= \frac{1}{14}\begin{bmatrix} 29 & -15 \\ -30 & 16 \end{bmatrix}$$

$= (AB)^{-1}$

Answer: $(AB)^{-1} = B^{-1}A^{-1}$ (verified)

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(vi) $AA^{-1} = A^{-1}A = I$

$$A^{-1} = \frac{-1}{2}\begin{bmatrix} 5 & -3 \\ -4 & 2 \end{bmatrix} = \begin{bmatrix} -5/2 & 3/2 \\ 2 & -1 \end{bmatrix}$$

$$AA^{-1} = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} \times \begin{bmatrix} -5/2 & 3/2 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A^{-1}A = \begin{bmatrix} -5/2 & 3/2 \\ 2 & -1 \end{bmatrix} \times \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Answer: $AA^{-1} = A^{-1}A = I$ (verified)

(vii) $(AB)^t = B^tA^t$

$$\text{L.H.S.} = (AB)^t = \begin{bmatrix} 16 & 30 \\ 15 & 29 \end{bmatrix}$$

$$B^t = \begin{bmatrix} 5 & 2 \\ 6 & 1 \end{bmatrix} \quad A^t = \begin{bmatrix} 2 & 4 \\ 3 & 5 \end{bmatrix}$$

$$\text{R.H.S.} = B^tA^t = \begin{bmatrix} 5 & 2 \\ 6 & 1 \end{bmatrix} \times \begin{bmatrix} 2 & 4 \\ 3 & 5 \end{bmatrix} = \begin{bmatrix} 16 & 30 \\ 15 & 29 \end{bmatrix}$$

Answer: L.H.S. = R.H.S. (verified)

(viii) (AB)C = A(BC)

$$BC = \begin{bmatrix} 5 & 6 \\ 2 & 1 \end{bmatrix} \times \begin{bmatrix} 3 & 2 \\ 1 & 7 \end{bmatrix} = \begin{bmatrix} 21 & 52 \\ 7 & 11 \end{bmatrix}$$

$$\text{L.H.S.} = \begin{bmatrix} 16 & 15 \\ 30 & 29 \end{bmatrix} \times \begin{bmatrix} 3 & 2 \\ 1 & 7 \end{bmatrix} = \begin{bmatrix} 63 & 137 \\ 119 & 263 \end{bmatrix}$$

$$\text{R.H.S.} = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} \times \begin{bmatrix} 21 & 52 \\ 7 & 11 \end{bmatrix} = \begin{bmatrix} 63 & 137 \\ 119 & 263 \end{bmatrix}$$

Answer: L.H.S. = R.H.S. (verified)

4. If A = [[7, 3], [2, 1]] and B = [[5, 3], [2, 4]], then find:

$$A = \begin{bmatrix} 7 & 3 \\ 2 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 5 & 3 \\ 2 & 4 \end{bmatrix}$$

(i) $|B|$ (ii) Adj B (iii) A^{-1}

(iv) $A^{-1}A$ (v) $(AB)^t$ (vi) $(B^t)^t$

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(i) |B|

$$|B| = \begin{vmatrix} 5 & 3 \\ 2 & 4 \end{vmatrix}$$

$= (5)(4) - (3)(2)$

$= 20 - 6$

Answer: |B| = 14

(ii) Adj B

Interchange 5 and 4, and change the signs of 3 and 2

$$\text{adj } B = \begin{bmatrix} 4 & -3 \\ -2 & 5 \end{bmatrix}$$

Answer: Adj B = [[4 −3], [−2 5]]

(iii) A^{-1}

$$|A| = \begin{vmatrix} 7 & 3 \\ 2 & 1 \end{vmatrix}$$

$= (7)(1) - (3)(2) = 7 - 6$

$= 1$

$$\text{adj } A = \begin{bmatrix} 1 & -3 \\ -2 & 7 \end{bmatrix}$$

$$A^{-1} = \frac{1}{1} \begin{bmatrix} 1 & -3 \\ -2 & 7 \end{bmatrix}$$

Answer: A^{-1} = [[1 −3], [−2 7]]

(iv) $A^{-1}A$

$$A^{-1}A = \begin{bmatrix} 1 & -3 \\ -2 & 7 \end{bmatrix} \times \begin{bmatrix} 7 & 3 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Answer: $A^{-1}A = I$

(v) $(AB)^t$

$$AB = \begin{bmatrix} 7 & 3 \\ 2 & 1 \end{bmatrix} \times \begin{bmatrix} 5 & 3 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 41 & 33 \\ 12 & 10 \end{bmatrix}$$

$$(AB)^t = \begin{bmatrix} 41 & 12 \\ 33 & 10 \end{bmatrix}$$

Answer: $(AB)^t$ = [[41 12], [33 10]]

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(vi) $(B^t)^t$

$$B^t = \begin{bmatrix} 5 & 2 \\ 3 & 4 \end{bmatrix}$$

$$(B^t)^t = \begin{bmatrix} 5 & 3 \\ 2 & 4 \end{bmatrix} = B$$

Answer: $(B^t)^t = B$ = [[5 3], [2 4]]

5. Use matrix inversion method and Cramer's rule to solve the following pairs of linear equations, if possible:

(i) $3x + 4y = 7,\ 5x - y = 2$

(ii) $x - 6y = -15,\ 2x + 6y = -3$

(iii) $2x + y = 5,\ x + 3y = 3$

(i) $3x + 4y = 7,\ 5x - y = 2$

$$\begin{bmatrix} 3 & 4 \\ 5 & -1 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 7 \\ 2 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 3 & 4 \\ 5 & -1 \end{vmatrix}$$

$= (3)(-1) - (4)(5) = -3 - 20$

$= -23$

$$\text{adj } A = \begin{bmatrix} -1 & -4 \\ -5 & 3 \end{bmatrix}$$

$$X = A^{-1}B = \frac{1}{-23}\begin{bmatrix} -1 & -4 \\ -5 & 3 \end{bmatrix} \times \begin{bmatrix} 7 \\ 2 \end{bmatrix}$$

$$= \frac{1}{-23}\begin{bmatrix} -7 - 8 \\ -35 + 6 \end{bmatrix}$$

$$= \frac{1}{-23}\begin{bmatrix} -15 \\ -29 \end{bmatrix} = \begin{bmatrix} 15/23 \\ 29/23 \end{bmatrix}$$

Answer: By matrix inversion method: x = 15/23, y = 29/23

$$D = \begin{vmatrix} 3 & 4 \\ 5 & -1 \end{vmatrix} = -23$$

$$D_x = \begin{vmatrix} 7 & 4 \\ 2 & -1 \end{vmatrix} = (7)(-1) - (4)(2) = -15$$

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$$D_y = \begin{vmatrix} 3 & 7 \\ 5 & 2 \end{vmatrix} = (3)(2) - (7)(5) = -29$$

$$x = \frac{D_x}{D} = \frac{-15}{-23} = \frac{15}{23}$$

$$y = \frac{D_y}{D} = \frac{-29}{-23} = \frac{29}{23}$$

Answer: x = 15/23, y = 29/23

(ii) x − 6y = −15, 2x + 6y = −3

$$\begin{bmatrix} 1 & -6 \\ 2 & 6 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -15 \\ -3 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 1 & -6 \\ 2 & 6 \end{vmatrix}$$

$= (1)(6) - (-6)(2) = 6 + 12$

$= 18$

$$\text{adj } A = \begin{bmatrix} 6 & 6 \\ -2 & 1 \end{bmatrix}$$

$$X = A^{-1}B = \frac{1}{18} \begin{bmatrix} 6 & 6 \\ -2 & 1 \end{bmatrix} \times \begin{bmatrix} -15 \\ -3 \end{bmatrix}$$

$$= \frac{1}{18} \begin{bmatrix} -108 \\ 27 \end{bmatrix}$$

$$= \frac{1}{18} \begin{bmatrix} -108 \\ 27 \end{bmatrix} = \begin{bmatrix} -6 \\ 3/2 \end{bmatrix}$$

Now by Cramer's rule

$$D = 18, \quad D_x = \begin{vmatrix} -15 & -6 \\ -3 & 6 \end{vmatrix} = -90 - 18 = -108$$

$$D_y = \begin{vmatrix} 1 & -15 \\ 2 & -3 \end{vmatrix} = -3 + 30 = 27$$

$$x = \frac{-108}{18} = -6, \quad y = \frac{27}{18} = \frac{3}{2}$$

Answer: x = −6, y = 3/2

(iii) 2x + y = 5, x + 3y = 3

$$\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

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$$|A| = \begin{vmatrix} 2 & 1 \\ 1 & 3 \end{vmatrix}$$

$= (2)(3) - (1)(1) = 6 - 1$

$= 5$

$$\text{adj } A = \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}$$

$$X = A^{-1}B = \frac{1}{5}\begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix} \times \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

$$= \frac{1}{5}\begin{bmatrix} 12 \\ 1 \end{bmatrix}$$

$$= \frac{1}{5}\begin{bmatrix} 12 \\ 1 \end{bmatrix} = \begin{bmatrix} 12/5 \\ 1/5 \end{bmatrix}$$

Now by Cramer's rule

$$D = 5, \quad D_x = \begin{vmatrix} 5 & 1 \\ 3 & 3 \end{vmatrix} = 15 - 3 = 12$$

$$D_y = \begin{vmatrix} 2 & 5 \\ 1 & 3 \end{vmatrix} = 6 - 5 = 1$$

$$x = \frac{12}{5}, \quad y = \frac{1}{5}$$

Answer: x = 12/5, y = 1/5

6. Find two numbers by using matrices such that twice the first added to the second makes 21, and twice the second added to the first makes 27.

Solution

Let the first number be x and the second number be y

Twice the first plus the second: $2x + y = 21$

Twice the second plus the first: $x + 2y = 27$

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 21 \\ 27 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix}$$

$= (2)(2) - (1)(1) = 4 - 1$

$= 3$

$$\text{adj A} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

$$X = A^{-1}B = \frac{1}{3}\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \times \begin{bmatrix} 21 \\ 27 \end{bmatrix}$$

$$= \frac{1}{3}\begin{bmatrix} 15 \\ 33 \end{bmatrix}$$

$$= \frac{1}{3}\begin{bmatrix} 15 \\ 33 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

Answer: The two numbers are 5 and 11

7. 4 knives and 6 forks cost Rs. 136, whereas 6 knives and 5 forks cost Rs. 164. Find the cost of a knife and a fork by using matrices.

Solution

Let the cost of one knife be x rupees and of one fork be y rupees

$4x + 6y = 136$

$6x + 5y = 164$

$$\begin{bmatrix} 4 & 6 \\ 6 & 5 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 136 \\ 164 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 4 & 6 \\ 6 & 5 \end{vmatrix}$$

$= (4)(5) - (6)(6) = 20 - 36$

$= -16$

$$\text{adj A} = \begin{bmatrix} 5 & -6 \\ -6 & 4 \end{bmatrix}$$

$$X = A^{-1}B = \frac{1}{-16}\begin{bmatrix} 5 & -6 \\ -6 & 4 \end{bmatrix} \times \begin{bmatrix} 136 \\ 164 \end{bmatrix}$$

$$= \frac{1}{-16}\begin{bmatrix} -304 \\ -160 \end{bmatrix}$$

$$= \frac{1}{-16}\begin{bmatrix} -304 \\ -160 \end{bmatrix}$$

$$= \begin{bmatrix} 19 \\ 10 \end{bmatrix}$$

Answer: Cost of a knife = Rs. 19, cost of a fork = Rs. 10

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8. A shop employs 5 men and 3 women and pays total daily wages Rs. 3500. If the number of men is reduced to 2 and 3 extra women are taken on, the daily wages amount to Rs. 5000. Find the daily wages of a man and a woman by using matrices.

Solution

Let the daily wage of a man be x rupees and of a woman be y rupees

First case: $5x + 3y = 3500$

Second case: men become 2 and women become $3 + 3 = 6$, so $2x + 6y = 5000$

$$\begin{bmatrix} 5 & 3 \\ 2 & 6 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3500 \\ 5000 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 5 & 3 \\ 2 & 6 \end{vmatrix}$$

$= (5)(6) - (3)(2) = 30 - 6$

$= 24$

$$\text{adj } A = \begin{bmatrix} 6 & -3 \\ -2 & 5 \end{bmatrix}$$

$$X = A^{-1}B = \frac{1}{24} \begin{bmatrix} 6 & -3 \\ -2 & 5 \end{bmatrix} \times \begin{bmatrix} 3500 \\ 5000 \end{bmatrix}$$

$$= \frac{1}{24} \begin{bmatrix} 6000 \\ 18000 \end{bmatrix}$$

$$= \frac{1}{24} \begin{bmatrix} 6000 \\ 18000 \end{bmatrix}$$

$$= \begin{bmatrix} 250 \\ 750 \end{bmatrix}$$

Answer: Daily wage of a man = Rs. 250, daily wage of a woman = Rs. 750

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